

Muhammad Rayyan

✉ mshakoori@tulane.edu —  LinkedIn —  GitHub —  Google Scholar

Research Interests: Medical AI, Multimodal Deep Learning, Medical Imaging, Generative AI

About

CS PhD student at Tulane University advancing Medical AI, Multimodal Deep Learning, and Generative AI across complex multimodal clinical data. Investigating medical world models to simulate disease trajectories, forecast future clinical states and diagnostic outcomes, and learn from medical imaging.

Education

Tulane University

PhD in Computer Science

Aug 2026 – Present

New Orleans, LA, USA

Old Dominion University

MS in Computer Science

Aug 2025 – May 2026

Norfolk, VA, USA

- **CGPA:** 4.00/4.00

National University of Sciences and Technology

BE in Software Engineering

Nov 2021 – May 2025

Islamabad, PK

- **CGPA:** 3.54/4.00
- **Thesis:** Analytica AI: Unified Cross Data Analytics via Multi Agent Systems

Publications

MCTS Guided Chunk Optimization with Beam Search for Retrieval Augmented Generation *IEEE-FIT'25*

S. Farooq, M Rayyan S, N. Iltaf, I. Islam · DOI: [10.1109/FIT67061.2025.11333696](https://doi.org/10.1109/FIT67061.2025.11333696)

[Published]

- Developed a structured retrieval pipeline for RAG using efficient chunking, semantic filtering, and duplicate passage suppression to optimize context relevance.

LLMs with Tools: An Empirical Evaluation of the Thought Action Observation Paradigm *ICMI'25*

M. Usman, N. Iltaf, M Rayyan S, T. A. Rana · DOI: [10.1109/ICMI68585.2026.11539909](https://doi.org/10.1109/ICMI68585.2026.11539909)

[Published]

- Empirically evaluated TAO based tool augmented LLMs, analyzing tool selection, parameter extraction, execution success, and response integration.

Multi Head Attention for Pulmonary Embolism Detection in CT Angiography *IEEE-IBCAST'25*

Z. T. Hussain, M Rayyan S, A. Gohar · DOI: [10.13140/RG.2.2.24206.88646](https://doi.org/10.13140/RG.2.2.24206.88646)

[Accepted]

- Proposed MHA UNet to enhance detection of peripheral emboli in CT pulmonary angiography, outperforming standard baselines such as nnU-Net and Swin Transformer.

Manuscripts Under Review

Foveated ViT: Pre-Encoder Token Sparsification for Efficient Plant Seedling Recognition

M Rayyan S, N. Adnan, S. Fang, F. Abbas, M. Umar

[Under review]

- Proposed Foveated ViT, utilizing a lightweight Scout Network to sparsify background tokens prior to self-attention layers, reaching 97.65% accuracy with 4.3M parameters.

A Survey of Methods, Architectures, and Deployment Challenges for Large Reasoning Models

M Rayyan S, Z. Qadir, A. Hamza, M. Quddus, R. Khan, S. Younas, M. Usman, N. Saeed

[Under review]

- Authored a comprehensive survey synthesizing core methodologies driving advanced LLM reasoning, covering reinforcement learning, search algorithms, and inference time compute scaling.

Multomics Reveals Epigenetic Control of Fibroblast Activity After Myocardial Infarction and a Key Role for RUNX Transcription Factors

*Y. Li, X. Zhang, K. Ghimire, N. Khatri, L. Wang, Q. Liu, M. Shiri, M Rayyan S, S. Upad-
hayaya, B. Li, J. Francis, C. A. Simintiras, R. Li, J. Sun, X. Fu*

[Under review]

- Contributed to computational and functional genomic analysis in a multomics cardiac repair investigation.

Myofibroblast Contractile Machinery is Essential for Cardiac Repair after Myocardial Infarction

L. Wang, Y. Li, X. Zhang, Q. Liu, K. Ghimire, N. Khatri, M Rayyan S, J. Jung, J. Sun, X. Fu [Under review]

- Contributed to data processing and computational analysis evaluating cardiac repair mechanisms post-myocardial infarction.

Privacy Preserving AI Agent Architecture for Medical Calculators

I. Itrat, M Rayyan S, S. Sheikh, N. Bibi [Under review]

- Developed a Model Context Protocol (MCP) and small language model (SLM) based agentic framework for secure, privacy preserving clinical calculations.

Manuscripts in Preparation

A Unified Multi Task Architecture for Joint Pancreatic Anatomy Lesion Segmentation and Tumor Grading

M Rayyan S, J. Hamm [In Preparation]

- Investigating unified multimodal volumetric architectures for simultaneous pancreatic & tumor segmentation and along with classification tasks

Dynamic Cardiovascular World Models for Adverse Event Forecasting from Longitudinal EHR Trajectories

M Rayyan S, J. Hamm, F. Han [In Preparation]

- Developing a time aware cardiovascular world model on large scale EHR trajectories to learn latent patient disease state transitions and dynamically predict major adverse cardiovascular events (MACE).

Research Experience

Research Assistant

Tulane University

Aug 2026 - Present

New Orleans, LA, USA

- Developing multimodal 3D CT and clinical pipelines predicting PNET survival, progression, and grade (nested 5×5 CV AUROCs: 0.869, 0.841, 0.662), integrating focal lesion segmentation achieving a 0.35 tumor Dice score; expanding to the PANT cohort for external validation.
- Investing time aware cardiovascular world models over large scale EHR trajectories to model latent disease state transitions and forecast prospective Major Adverse Cardiovascular Events (MACE).

Research Assistant

Old Dominion University

Aug 2025 - May 2026

Norfolk, VA, USA

- Generated large scale custom genomic datasets and benchmarked Sei, Enformer, and ChromBPNet architectures for predicting 21,907 chromatin profiles, reaching up to 0.977 AUROC for SEI, followed by 0.9437 and 0.9039 by Enformer and ChromBPNET.
- Explored DNA foundation models including DNABERT, SPACE, and JanusDNA for predicting functional genomic events.

Research Assistant

National University of Sciences and Technology

Jul 2024 - May 2025

Islamabad, PK

- Worked on "Judgement Augmented Text-to-SQL via Natural Language Interfaces," where benchmarked instruction tuned models. Further enhanced reliability by integrating a Llama 3.1 405B judge model for self correction, drastically reducing hallucinations.
- Worked on "A Multi Agent System for Unified Analytics over Heterogeneous Data Sources". The final LangGraph orchestrated system generates analytical reports and visualizations to enhance strategic decision making.

Internships

Software Engineer Intern

Telenor Microfinance Bank

Jul 2023 - Aug 2023

Islamabad, PK

- Worked on an e commerce website using Java Spring Boot and Thymeleaf, implemented core functionalities with authentication features.

Achievements

- My team, “Kepler’s Horizon”, won the award for Best Use of Data and was nominated for Global Judging at the NASA Space Apps Challenge 2024.
- Final Year Project Thesis received an award from Chief Guest out of 64 total projects.
- Selected as one of 1,000 students globally for an AWS sponsored Scholarship for the AI Programming with Python Nanodegree course on Udacity.

Projects

AstroGPT-1.51B | Link: [git/astrogpt](https://github.com/astrogpt)

Tools: Python, LLM, LORA/QLORA, HuggingFace, Qwen model

Fine-tuned Qwen 2-1.5B using QLoRA on 22K tokens of my handwritten astrophysics articles for improved specialized scientific text generation.

ExCNET: Resource Efficient Attention Network for Seedling Recognition

Tools: Python, PyTorch, Scikit-Learn

Proposed ExCNET, an 8M parameter attention based architecture achieving 97% test accuracy on plant seedling recognition, outperforming heavier models including ResNet-50/152V2, VGG-16/19, DenseNet169, and ViT.

2D Multiple Objects Detection and Tracking for Autonomous Vehicles | Link: [git/auto](https://github.com/astrogpt)

Tools: Python, Ultralytics, YOLO, Matplotlib

Trained various YOLO models on the KITTI dataset, achieving high accuracy with YOLOv9t with 0.9798 precision and 0.7846 recall over YOLO v8n, 10n and 11n.

ML for Splice Site Identification in gene expression | Link: [git/gene](https://github.com/astrogpt)

Tools: Python, Numpy, Pandas, Scikit-learn, Tensorflow

Systematically trained and evaluated 13 ML/DL architectures. Achieving a top weighted F1 of 0.974 with an optimized SVM, highlighting increased model complexity does not guarantee superior performance.

Skills

Languages: Python, JavaScript, C++, SQL

Frameworks: LangChain, LangGraph, CrewAI, Streamlit, React

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, PyTorch, TensorFlow, Nltk, OpenCV, Yolo

Tools: Jupyter Notebook, Google Colab, Kaggle notebooks, PyCharm, Visual Studio Code

Extracurricular Activities

Authored Articles on Machine Learning

Published Medium articles on ML algorithms, such as Linear Regression, Naïve Bayes, KNN, and Logistic Regression, simplifying complex concepts for beginners with practical Python implementations. [[Medium](#)]

Memberships

- Professional Member, IEEE

Apr 2026 – Present

Academic Service

- Peer Reviewer for IEEE Transactions on Systems, Man, and Cybernetics
- Peer Reviewer for IEEE Access
- Peer Reviewer for SciPy Conference 2026